

SMARTSPINNER v0.4

PCB engineering completion and turnkey PCBA request for quotation

100 mm four-arm persistence-of-vision display with BLE



100 mm	outer diameter
4	main arms
128	RGB pixels total
10 sets	requested lot

RELEASE STATUS: ENGINEERING PACKAGE - NOT FOR FABRICATION

The supplied design is intended for supplier engineering review, layout completion, DFM and quotation. Fabrication may start only after the corrected editable KiCad project, final outputs and zero-error DRC reports have been approved in writing.

Prepared for supplier review | Project owner: German Klimenko

1. Product and electronics architecture

SmartSpinner is a hand-spun POV display. Radial LEDs form a circular image while vertical tip boards form a cylindrical ticker around the outer edge.



One electronic set

- 1 four-arm main PCB with 80 top SK6805-EC15 RGB pixels
- 4 identical tip PCBs with 12 SK6805-EC10 RGB pixels each
- 8 independently controlled LED data streams
- nRF52840 BLE, BMI270 IMU and DRV5033 Hall index sensor
- Protected 1S LiPo input, 3.3 V logic and switched 5 V LED rail

Mechanical envelope

- 100 mm outer diameter; four symmetric arms; 608 center bearing
- Target enclosure thickness approximately 16 mm in v0.3 mechanics
- Main PCB up to approximately 96 mm envelope, target 1.0 mm four-layer construction
- Tip PCB target thickness approximately 0.6 mm; supplier to confirm manufacturability

Critical design principle

Heavy components and batteries remain close to the center. Mass distribution must stay four-way symmetric. Component substitutions and copper changes must not create a meaningful imbalance between arms.

2. Requested supplier engineering work

Step	Deliverable	Required work
01	Audit	Review the architecture, supplied PCB sources, DRC reports, power budget and mechanical constraints.
02	Complete	Close every open connection and complete charge, boost, switching, protection and decoupling details.
03	Verify	Check exact MPN footprints, pin-1 orientation, RF keep-out, stack-up, clearances and sourcing.
04	Return	Provide editable KiCad sources, zero-error DRC, final Gerber/drill, BOM, CPL, drawings and DFM report.

Known status

Main PCB	68 unconnected items
Tip PCB	4 unconnected items
Gerbers	Review-only; regenerate after design approval
Footprints	Verify against exact selected manufacturer parts

Fabrication gate

Both board types must reach zero copper errors and zero unconnected items. Every remaining ERC/DRC waiver and DFM deviation must be documented and accepted before purchase approval.

3. Quotation and first-article plan

Please quote engineering and manufacturing as separate commercial stages.

Stage	Requested quantity / result	Supplier response
Engineering	Editable KiCad, DRC=0, final outputs, DFM	Price + lead time + revision rounds
First article	1 main PCB + 4 tip PCBs, assembled and tested	Price + lead time + test coverage
Remaining lot	9 main PCBs + 36 tip PCBs	Price + lead time after approval
Alternatives	5, 10 and 25 complete-set breaks	Unit and total prices
Optional	Programming and functional test	NRE + unit price
Shipping	Moscow, Russia; no LiPo batteries	Method + price + delivery estimate

Required first-article checks

- AOI and visual inspection
- Battery, 3.3 V and 5 V rail short-circuit checks
- Charging, boost and switched LED rail verification with laboratory current limiting
- MCU programming, BLE, Hall and IMU verification
- All eight LED streams tested at limited brightness
- Current, temperature, mechanical fit and static balance recorded

Approval rule

Do not assemble the remaining units until German Klimenko has reviewed the revised design files and accepted the first-article results in writing.

Project website: <https://germanklimenko.github.io/smarts spinner/>

Repository: <https://github.com/GermanKlimenko/smarts spinner>